

YORUBA CALCULATOR PRO

Yoruba Number Grammar

A Modern Standardized System

DEVELOPED BY

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The official grammar of the Yoruba number system implemented in the Yoruba Calculator Pro app.

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1 Introduction

Yoruba is a language spoken by over 40 million people, yet its number system has historically lacked a **single, consistent, and scalable standard** for expressing very large numbers. Regional variations, subtractive patterns, and irregular forms have made it difficult for learners, teachers, and technology to handle Yoruba numerals with precision.

Yoruba Calculator Pro introduces a **modern, standardized Yoruba number system** that is:

- **Systematic** — built from a small set of rules
- **Additive and multiplicative** — reflecting the logic of the language
- **Scalable** — works up to 999,999,999,999,999,999,999
- **Digital-first** — designed for calculators, apps, and education

> **NOTE** This document is the **official grammar** of that system.

2 Core Principles of the System

The system rests on four principles:

Principle	Meaning	Example
Additive	Numbers are built by adding components	**ogún-ẹ̀ta** = 20 + 3 = 23
Multiplicative	Counters multiply units	**ọgọ́rùn-ún mẹ́ta** = 100 × 3 = 300
Hierarchical	Larger units come first, remainders after	**ẹgbẹ̀rún kan ọgọ́rùn-ún méjì** = 1,000 + 200
Recursive	The same rules apply at every magnitude	12, 1,200, 1,200,000 all follow the same pattern

3 The Building Blocks

The entire system is built from five sets of words.

A. Basic Numbers (0–19) — Unique Names

Number	Yoruba	Number	Yoruba
0	**odo**	10	**ẹ̀wá**
1	**ení**	11	**ọ̀kànlá**
2	**èjì**	12	**èjìlá**
3	**ẹ̀ta**	13	**ẹ̀tàlá**
4	**ẹ̀rin**	14	**ẹ̀rìnlá**
5	**àrún**	15	**àrùndógún**
6	**ẹ̀fà**	16	**ẹ̀rìndógún**
7	**èje**	17	**ẹ̀tàdógún**
8	**ẹ̀jọ**	18	**èjìdógún**
9	**ẹ̀sán**	19	**ọ̀kàndógún**

B. Tens (20–90) — Base Multipliers

Number	Yoruba
20	**ogún**
30	**ọgbọ̀n**
40	**ogójì**
50	**àádọ́ta**
60	**ọgọ́ta**
70	**àádọ́rin**
80	**ọgọ́rin**
90	**àádọ́rùn-ún**

C. Connectors (1–9) — Added to Tens

Unit	Connector	Unit	Connector	Unit	Connector
1	**-ení**	4	**-ẹ̀rin**	7	**-èje**
2	**-èjì**	5	**-àrún**	8	**-ẹ̀jọ**
3	**-ẹ̀ta**	6	**-ẹ̀fà**	9	**-ẹ̀sán**

D. Large Units — Multipliers for Big Numbers

Power	Yoruba	Value
100	**ọgọ́rùn-ún**	10^2
1,000	**ẹgbẹ̀rún**	10^3
1,000,000	**ẹgbẹ̀rúngbẹ̀rún**	10^6
1,000,000,000	**ẹ̀ṣú**	10^9
1,000,000,000,000	**ẹgbẹ̀rúnṣú**	10^{12}
1,000,000,000,000,000	**ẹgbẹ̀rúngbẹ̀rúnṣú**	10^{15}
1,000,000,000,000,000,000	**yẹ̀pẹ̀**	10^{18}

E. Counters — Multiply Large Units

Counter	Yoruba	Counter	Yoruba
1	**kan**	11	**mọ́kànlá**
2	**méjì**	12	**méjìlá**
3	**mẹ́ta**	13	**mẹ́tàlá**
4	**mẹ́rin**	14	**mẹ́rìnlá**
5	**márùn-ún**	15	**márùndógún**
6	**mẹ́fà**	16	**mẹ́rìndógún**
7	**méje**	17	**mẹ́tàdógún**
8	**mẹ́jọ**	18	**méjìdógún**
9	**mẹ́sàn-án**	19	**mọ́kàndógún**
10	**mẹ́wàá**		

4 Rule 1: Basic Numbers (0–19)

> **RULE** Numbers 0–19 have **unique names**. They are never built from smaller parts.

Number	Yoruba
0	**odo**
1	**ení**
2	**èjì**
…	…
19	**ọ̀kàndógún**

Note: These are the atoms of the system. Everything else is built from them.

5 Rule 2: Tens (20–90)

> **RULE** Tens have unique names. They are the **base multipliers** for compound numbers.

Number	Yoruba
20	**ogún**
30	**ọgbọ̀n**
40	**ogójì**
50	**àádọ́ta**
60	**ọgọ́ta**
70	**àádọ́rin**
80	**ọgọ́rin**
90	**àádọ́rùn-ún**

6 Rule 3: Compound Numbers (21–99)

> **RULE** Compound numbers = **Tens + Connector**

Formula

> **Compound = Tens word + Connector (with hyphen)**

Examples

Number	Yoruba	Breakdown
21	**ogún-ení**	20 + 1
22	**ogún-èjì**	20 + 2
25	**ogún-àrún**	20 + 5
32	**ọgbọ̀n-èjì**	30 + 2
45	**ogójì-àrún**	40 + 5
57	**àádọ́ta-èje**	50 + 7
68	**ọgọ́ta-èjọ**	60 + 8
79	**àádọ́rin-ẹ̀sán**	70 + 9
84	**ọgọ́rin-ẹ̀rin**	80 + 4
99	**àádọ́rùn-ún-ẹ̀sán**	90 + 9

NOTE The hyphen is compulsory. It shows that the connector belongs to the tens word.

7 Rule 4: Hundreds (100–900)

RULE Hundreds = **ọgọ́rùn-ún** + **Counter**

Formula

Hundreds = ọgọ́rùn-ún + Counter

Examples

Number	Yoruba	Breakdown
100	**ọgọ́rùn-ún kan**	100 × 1
200	**ọgọ́rùn-ún méjì**	100 × 2
300	**ọgọ́rùn-ún mẹ́ta**	100 × 3
400	**ọgọ́rùn-ún mẹ́rin**	100 × 4
500	**ọgọ́rùn-ún márùn-ún**	100 × 5
600	**ọgọ́rùn-ún mẹ́fà**	100 × 6
700	**ọgọ́rùn-ún méje**	100 × 7
800	**ọgọ́rùn-ún mẹ́jọ**	100 × 8
900	**ọgọ́rùn-ún mẹ́sàn-án**	100 × 9

8 Rule 5: Thousands (1,000–999,000)

> **RULE** Thousands = **ẹgbẹ̀rún** + **Counter**

Number	Yoruba	Breakdown
1,000	**ẹgbẹ̀rún kan**	1,000 × 1
2,000	**ẹgbẹ̀rún méjì**	1,000 × 2
3,000	**ẹgbẹ̀rún mẹ́ta**	1,000 × 3
10,000	**ẹgbẹ̀rún mẹ́wàá**	1,000 × 10
15,000	**ẹgbẹ̀rún márùndógún**	1,000 × 15
20,000	**ogún ẹgbẹ̀rún**	20 × 1,000
30,000	**ọgbọ̀n ẹgbẹ̀rún**	30 × 1,000
45,000	**ogójì-àrún ẹgbẹ̀rún**	45 × 1,000
99,000	**àádọ́rùn-ún-ẹ̀sán ẹgbẹ̀rún**	99 × 1,000
100,000	**ọgọ́rùn-ún kan ẹgbẹ̀rún**	100 × 1,000
200,000	**ọgọ́rùn-ún méjì ẹgbẹ̀rún**	200 × 1,000
500,000	**ọgọ́rùn-ún márùn-ún ẹgbẹ̀rún**	500 × 1,000

> **NOTE** For numbers 20,000 and above, the counter is itself a compound or simple number. The pattern is: **Counter** + **ẹgbẹ̀rún**.

9 Rule 6: Millions and Beyond

The same pattern repeats for every large unit.

Millions (10^6)

Number	Yoruba
1,000,000	**ẹgbẹ̀rúngbẹ̀rún kan**
2,000,000	**ẹgbẹ̀rúngbẹ̀rún méjì**
20,000,000	**ogún ẹgbẹ̀rúngbẹ̀rún**
200,000,000	**ọgọ́rùn-ún méjì ẹgbẹ̀rúngbẹ̀rún**

Billions (10^9)

Number	Yoruba
1,000,000,000	**ẹ̀ṣú kan**
2,000,000,000	**ẹ̀ṣú méjì**
20,000,000,000	**ogún ẹ̀ṣú**
200,000,000,000	**ọgọ́rùn-ún méjì ẹ̀ṣú**

Trillions (10^{12})

Number	Yoruba
1,000,000,000,000	**ẹgbẹ̀rúnṣú kan**
20,000,000,000,000	**ogún ẹgbẹ̀rúnṣú**

Quadrillions (10^{15})

Number	Yoruba
1,000,000,000,000,000	**ẹgbẹ̀rúngbẹ̀rúnṣú kan**
20,000,000,000,000,000	**ogún ẹgbẹ̀rúngbẹ̀rúnṣú**

Quintillions (10^{18})

Number	Yoruba
1,000,000,000,000,000,000	**yẹ̀pẹ̀ kan**
20,000,000,000,000,000,000	**ogún yẹ̀pẹ̀**

10 Rule 7: The Remainder Rule

> **RULE** After stating the largest unit, add the remainder using the same rules.

Formula

> **Full number = [Largest unit part] + [Remainder]**

Examples

Number	Yoruba	Breakdown
108	**ọgọ́rùn-ún kan ẹ̀jọ**	100 + 8
205	**ọgọ́rùn-ún méjì àrún**	200 + 5
321	**ọgọ́rùn-ún mẹ́ta ogún-ení**	300 + 21
1,234	**ẹgbẹ̀rún kan ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin**	1,000 + 200 + 30 + 4
2,500	**ẹgbẹ̀rún méjì ọgọ́rùn-ún márùn-ún**	2,000 + 500
12,345	**ẹgbẹ̀rún méjìlá ọgọ́rùn-ún mẹ́ta ogójì-àrún**	12,000 + 300 + 45
1,234,567	**ẹgbẹ̀rúngbẹ̀rún kan ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin ẹgbẹ̀rún ọgọ́rùn-ún márùn-ún ọgọ́ta èje**	1,000,000 + 234,000 + 567

11 Complete Examples

Example 1: 25

> **ogún-àrún**

20 + 5 · Compound number

Example 2: 200

> **ọgọ́rùn-ún méjì**

100 × 2 · Hundred with counter

Example 3: 108

> **ọgọ́rùn-ún kan ẹ̀jọ**

100 + 8 · Hundred + remainder

Example 4: 20,000

> **ogún ẹgbẹ̀rún**

20 × 1,000 · Counter + thousand

Example 5: 200,300

> **ọgọ́rùn-ún méjì ẹgbẹ̀rún ọgọ́rùn-ún mẹ́ta**

200 × 1,000 + 300 · Counter + thousand + remainder

Example 6: 1,234,567

> **ẹgbẹ̀rúngbẹ̀rún kan ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin ẹgbẹ̀rún ọgọ́rùn-ún márùn-ún ọgọ́ta èje**

1,000,000 + 234,000 + 567 · Million + thousand + remainder

Example 7: 12,345,678,901

> **ẹ̀ṣú méjìlá ọgọ́rùn-ún mẹ́ta ogójì-àrún ẹgbẹ̀rúngbẹ̀rún ọgọ́rùn-ún mẹ́fà àádọ́rin-ẹ̀jọ ẹgbẹ̀rún ọgọ́rùn-ún mẹ́sàn-án kan**

12,000,000,000 + 345,000,000 + 678,000 + 901 · Billion + million + thousand + remainder

12 Pronunciation and Gap Guide

When speaking Yoruba numbers, the **gaps between words** follow the mathematical structure.

Gap Type	Duration	When to Use	Example
NO_GAP	0 ms	Within compounds, unit + counter	**ogún-ení**, **ọgọ́rùn-ún méjì**
TINY_GAP	0 ms	Counter + unit	**ogún ẹgbẹ̀rún**
NORMAL_GAP	200 ms	Between major components	**ẹgbẹ̀rún kan** + pause + **ọgọ́rùn-ún méjì**

Visual Guide

> **25**
>
> **ogún-àrún**
>
> **[NO GAP]** — sounds like one word

> **200**
>
> **ọgọ́rùn-ún méjì**
>
> **[NO GAP]** — sounds like one phrase

> **108**
>
> **ọgọ́rùn-ún kan ẹ̀jọ**
>
> **[NO GAP]** within the hundred, then **[NORMAL GAP]** before the remainder

20,000

ogún ẹgbẹ̀rún

[TINY GAP] between counter and unit

13 Comparison with Traditional Counting

Number	Traditional (Subtractive)	Modern (This System)
45	**Àrùndínláàádọ́ta**	**ogójì-àrún**
75	**àrùndínlọ́gọ́rin**	**àádọ́rin-àrún**
95	**àrùndínlọ́gọ́rùn-ún**	**àádọ́rùn-ún-àrún**
1,234	Often inconsistent	**ẹgbẹ̀rún kan ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin**
1,000,000	No common term	**ẹgbẹ̀rúngbẹ̀rún kan**

NOTE This system does **not replace** traditional counting. It offers a **standardized alternative** for education, technology, and large numbers.

14 Glossary of Terms

Term	Meaning	Example
Basic number	Unique name for 0–19	**ení**, **èjì**
Tens word	Base multiplier (20–90)	**ogún**, **ọgbọ̀n**
Connector	Unit added to tens	**-ení**, **-èjì**
Large unit	Multiplier for big numbers	**ọgọ́rùn-ún**, **ẹgbẹ̀rún**
Counter	Number that multiplies a unit	**kan**, **méjì**
Compound number	Tens + connector	**ogún-ení**
Remainder	What is left after a unit	8 in 108
NO_GAP	No pause between words	Within compounds
NORMAL_GAP	200 ms pause	Between components

15 Practice Exercises

Write each number in Yoruba using the rules above. Answers follow the exercises.

Exercise	Write in Yoruba
Exercise 1	15 · 28 · 47 · 63 · 81
Exercise 2	100 · 200 · 500 · 900 · 1,000
Exercise 3	108 · 234 · 567 · 1,234 · 12,345
Exercise 4	20,000 · 200,000 · 1,000,000 · 12,000,000 · 123,456,789
Exercise 5	1,000,000,000 · 12,345,678,901 · 999,999,999,999,999,999,999

Answer Key

Exercise 1

#	Number	Yoruba
1	15	**àrùndógún**
2	28	**ogún-ẹ̀jọ**
3	47	**ogójì-èje**
4	63	**ọgọ́ta-ẹ̀ta**
5	81	**ọgọ́rin-ení**

Exercise 2

#	Number	Yoruba
1	100	**ọgọ́rùn-ún kan**
2	200	**ọgọ́rùn-ún méjì**
3	500	**ọgọ́rùn-ún márùn-ún**
4	900	**ọgọ́rùn-ún mẹ́sàn-án**
5	1,000	**ẹgbẹ̀rún kan**

Exercise 3

#	Number	Yoruba
1	108	**ọgọ́rùn-ún kan ẹ̀jọ**
2	234	**ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin**
3	567	**ọgọ́rùn-ún márùn-ún ọgọ́ta-èje**
4	1,234	**ẹgbẹ̀rún kan ọgọ́rùn-ún méjì ọgbọ̀n-ẹ̀rin**
5	12,345	**ẹgbẹ̀rún méjìlá ọgọ́rùn-ún mẹ́ta ogójì-àrún**

Exercise 4

#	Number	Yoruba
1	20,000	**ogún ẹgbẹ̀rún**
2	200,000	**ọgọ́rùn-ún méjì ẹgbẹ̀rún**
3	1,000,000	**ẹgbẹ̀rúngbẹ̀rún kan**
4	12,000,000	**ẹgbẹ̀rúngbẹ̀rún méjìlá**
5	123,456,789	**ọgọ́rùn-ún kan ogún-ẹ̀ta ẹgbẹ̀rúngbẹ̀rún ọgọ́rùn-ún mẹ́rin àádọ́ta-ẹ̀fà ẹgbẹ̀rún ọgọ́rùn-ún méje ọgọ́rin-ẹ̀sán**

Exercise 5

#	Number	Yoruba
1	1,000,000,000	**ẹ̀ṣú kan**
2	12,345,678,901	**ẹ̀ṣú méjìlá ọgọ́rùn-ún mẹ́ta ogójì-àrún ẹgbẹ̀rúngbẹ̀rún ọgọ́rùn-ún mẹ́fà àádọ́rin-ẹ̀jọ ẹgbẹ̀rún ọgọ́rùn-ún mẹ́sàn-án kan**
3	999,999,999,999,999,999,999	**ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán yẹ̀pẹ̀ ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán ẹgbẹ̀rúngbẹ̀rúnṣú ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán ẹgbẹ̀rúnṣú ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán ẹ̀ṣú ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán ẹgbẹ̀rúngbẹ̀rún ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán ẹgbẹ̀rún ọgọ́rùn-ún mẹ́sàn-án àádọ́rùn-ún-ẹ̀sán**

> **TIP** Large numbers are converted recursively: split the number into groups of three digits, name each group using the hundreds and compound rules, then attach the correct large unit.

YORUBA CALCULATOR PRO

Speak Yoruba Numbers Fluently

ABOUT THE BOOK

Yoruba Number Grammar sets out a modern, standardized system for reading and writing Yoruba numbers, from 0 up to 999 quintillion. Built on seven clear rules that are additive, multiplicative, hierarchical and recursive, it covers basic numbers, tens, compounds, hundreds, thousands, large units and remainders. It also includes worked examples, a pronunciation and gap guide, a comparison with traditional counting, a glossary, and practice exercises with answers.



ABOUT YORUBA CALCULATOR PRO

A scientific Android calculator that is very robust and speaks everything: every number you type and every result, however complex, is read aloud in Yoruba. It is the reference implementation of the grammar in this book, and a practical way to learn and teach Yoruba numerals.

GET IT ON GOOGLE PLAY
https://play.google.com/store/apps/details?id=com.hakeemoadeniyan.yorubacalculatorpro



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